Constructing and Analyzing a Standarized IT Recruitment Dataset via LLM-Assisted Extraction and Statistical Imputation

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***Abstract*—Data preprocessing and dataset construction are foundational components in data science, directly governing the integrity of downstream analytical models and statistical insights. Although online job platforms contain large volumes of recruitment data, job descriptions are frequently unstructured, inconsistent, and non-standardized. In this paper, we present a principle-driven data architecture and preprocessing pipeline designed to construct a structured dataset from heterogeneous IT recruitment texts in Vietnam. A raw dataset of 250 job postings is collected while strictly preserving data provenance under the Immutable Raw Principle. A 100-sample subset is then manually annotated through independent cross-labeling and a formal conflict resolution framework to construct a verified Ground Truth. We leverage a Large Language Model configured in fully deterministic mode (Temperature = 0.0) to execute zero-shot information extraction on the remaining 150 records, achieving a Cohen's Kappa above 0.83 and a Jaccard similarity above 76%, indicating strong agreement with human annotation. Salary data is normalized into uniform monthly numeric boundaries, and missing values are handled through a Groupby Median imputation strategy under the Missing at Random assumption. Finally, an Exploratory Data Analysis reveals localized salary benchmarks and technical skill co-occurrence patterns across job domains. This work delivers an analysis-ready dataset that serves as a reproducible foundation for future labor market research in the Vietnamese IT sector. The dataset and full preprocessing codebase are publicly available at: https://github.com/phonghoangKHDL/IT_Recruitment_Preprocessing.**

***Index Terms*—Data Preprocessing, Dataset Construction, Information Extraction, Large Language Models, Labor Market Analysis**

I. Introduction

The rapid growth of the Information Technology (IT) sector in Vietnam has created an increasing need for accurate and reliable market intelligence to guide workforce development and recruitment strategy [1]. Although major online recruitment platforms such as ITviec and TopDev host large volumes of job postings, most actionable information remains embedded in free-text job descriptions that are unstructured and difficult to process at scale. Extracting key entities such as technical skills, salary ranges, experience requirements, and language qualifications from these texts presents significant technical challenges. Manual analysis is infeasible at scale, while rule-based approaches such as regular expressions frequently fail in the face of semantic diversity and formatting inconsistency across real-world postings.

The emergence of Large Language Models (LLMs) has opened a promising direction for automated structuring of unstructured text. However, most existing work focuses on downstream applications while investing insufficient effort in the foundational layer: the rigorous and reproducible construction of clean, structured datasets. Without a carefully controlled preprocessing architecture, downstream models risk inheriting statistical bias or hallucination artifacts from noisy source data.

To address this gap, this paper presents a principle-driven data architecture and preprocessing pipeline designed to transform unstructured IT recruitment texts into a structured, analysis-ready dataset for the Vietnamese IT labor market. Rather than building predictive models, this study focuses on the rigor of the data construction lifecycle itself. A semi-automated pipeline is designed to combine human validation with the zero-shot extraction capability of an LLM. By adhering to the Immutable Raw Principle from the collection stage and establishing a verified Ground Truth through independent cross-labeling, the pipeline produces a reliable reference standard for evaluating AI extraction quality prior to full automation.

The main contributions of this paper are as follows:

- **A principle-driven LLM extraction framework:** A data architecture integrating Prompt Engineering with an LLM operating in deterministic mode (Temperature = 0.0), validated against a human-annotated Ground Truth and achieving a Cohen's Kappa above 0.83 and Jaccard similarity above 76%.
- **A MAR-aware statistical imputation strategy:** A two-tier Experience-Aware Nearest Neighbor imputation approach under the Missing at Random (MAR) assumption, designed specifically for the high rate of undisclosed salary data in Vietnamese recruitment postings.

- **A structured, analysis-ready dataset:** A standardized collection of 250 IT recruitment records with harmonized geographic classifications, structured monthly salary boundaries in VND, and skill co-occurrence analysis across job domains.
- **A documented preprocessing reference:** Full transparency in pipeline design and methodology, providing a reproducible foundation for future labor market studies in the Vietnamese IT sector. The complete dataset and source code are open-sourced at https://github.com/phonghoangKHDL/IT_Recruitment_Preprocessing.

II. Related Work

A. Job Market Analysis and Skill Demand Studies

The analysis of labor market dynamics has traditionally relied on structured data sources such as government census reports. However, the proliferation of online recruitment platforms has shifted research attention toward processing free-text Job Descriptions (JDs). Recent studies highlight a critical gap in reliable, real-time career guidance within the Vietnamese IT job market, where existing market reports quickly become outdated [1]. Automated pipelines are increasingly required to replace manual analysis of job postings, which is impractical at scale for both researchers and practitioners [1]. These limitations underscore the need for a robust methodological foundation to convert raw, unstructured market data into analysis-ready datasets.

B. Information Extraction from Job Descriptions

Extracting structured information such as technical skills, experience requirements, and salary ranges from job postings is fundamentally an Information Extraction (IE) task. Historically, this was tackled using rule-based frameworks such as Regular Expressions, or traditional Named Entity Recognition (NER) models employing sequence labeling with BIO tags [2]. However, these approaches rely heavily on manually annotated training data and often struggle with syntactically complex skill mentions or ambiguous phrasing [2]. They also fail to handle linguistic diversity and irregular formatting across postings, particularly for nuanced fields such as compensation structures or inferred work arrangements [3].

C. Large Language Models for Structured Data Extraction

Large Language Models (LLMs) have significantly advanced the state of Information Extraction by enabling in-context and zero-shot generalization without extensive supervised fine-tuning. Recent work demonstrates superior handling of complex and ambiguous skill patterns compared to traditional NER approaches [2]. In labor market analytics, LLM-based and Retrieval-Augmented Generation (RAG) pipelines have shown strong performance in extracting job features from unstructured text [3], and have been deployed specifically for the Vietnamese market to structure raw posting data for real-time querying [1].

D. Data Quality and Missing Value Imputation

Despite the growing adoption of LLMs for data extraction, a significant gap remains in the methodological rigor of dataset construction and quality control. Many applications ingest scraped web data without strict provenance tracking, increasing the risk of hallucination artifacts propagating into downstream analyses. Establishing a validated Ground Truth through inter-annotator agreement metrics such as Cohen's Kappa [4] and Jaccard similarity [5] is therefore essential for scientific credibility. Handling incomplete records also requires principled strategies; statistical imputation methods tailored for Missing at Random (MAR) distributions, such as Groupby Median approaches, provide a transparent and defensible alternative to listwise deletion [6].

Despite these advances, existing studies rarely provide a fully documented preprocessing architecture that integrates deterministic LLM extraction, human-validated annotation, and statistically grounded missing-value handling within a single reproducible pipeline for the Vietnamese IT labor market. This work directly addresses that gap.

III. Methodology

To transform heterogeneous, unstructured job advertisements into a clean, structured, and analysis-ready resource, we design a rigorous, five-phase data preprocessing pipeline. The overall architectural workflow and data transitions of the proposed system are illustrated in Fig. 1, demonstrating the systematic progression from raw data acquisition to final exploratory analysis while maintaining strict data quality control and provenance tracking at every stage.

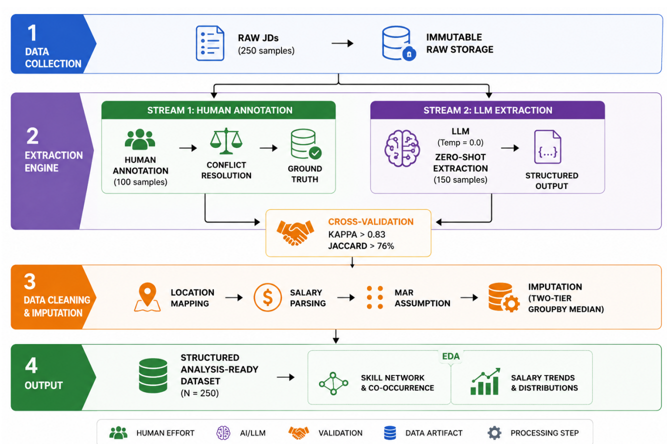


Fig. 1: The proposed principle-driven data preprocessing and dataset construction pipeline, detailing the integration of human validation, LLM-assisted extraction, and statistical imputation.

A. Phase 1: Raw Data Acquisition and Provenance

The pipeline initiates with the acquisition of $N = 250$ IT job postings from prominent Vietnamese recruitment platforms (ITviec and TopDev). Rather than deploying automated web scraping algorithms, we deliberately opted for a semi-manual data collection protocol. This architectural decision is grounded in the restrictive security infrastructures of modern job portals, which actively deploy strict anti-bot mechanisms (e.g., Cloudflare) and enforce authenticated sessions to access critical information such as salary ranges. If automated scraping tools were utilized without complex bypassing architectures, they would frequently encounter connection resets or fail to penetrate login walls. This would result in the systematic omission of crucial attributes, thereby introducing severe missing values (Missing Not at Random - MNAR) into the dataset from the very beginning.

Furthermore, raw job description text was collected by directly capturing the browser-rendered output rather than parsing raw HTML. This approach preserves the original plain-text structure of each posting, including structural line breaks and list formatting, which are frequently lost during standard HTML tag removal by automated scrapers. Consequently, the true unstructured yet spatially organized nature of the labor market data is faithfully captured.

The schema of the initial uncleaned dataset (`Data_Raw.csv`) is detailed in Table 1. It consists of six primary fields, each exhibiting high volatility and unique noise characteristics that necessitate advanced downstream preprocessing.

Raw Field	Description and Noise Characteristics
`Job_Title`	Job position title as posted by the employer.
`Company`	Employer name.
`Location`	Workplace location as entered by the employer. Highly unstandardized, with inconsistent representations of the same city or district across postings.
`Salary_Raw`	Raw salary string in its original form. Contains multiple currencies (USD, VND), non-numeric disclosures such as ”Negotiable”, and inconsistent range formats.
`Link`	Source URL of the job posting, retained for provenance tracking.
`JD & Requirements`	Full job description text as provided by the employer. Unstructured and multilingual (Vietnamese-English), with inconsistent formatting across postings.

TABLE 1: Raw Dataset Schema and Noise Characteristics

At this stage, the *Immutable Raw Principle* is strictly enforced. No programmatic cleaning—such as whitespace stripping, lowercasing, or regex-based filtering—was permitted during ingestion. If standard text normalization were applied prematurely at this phase, it would irreversibly erase the contextual boundaries of sensitive technical entities (e.g., conflating ”C++” with ”C”, or ”.NET” with ”net”), thereby introducing severe semantic biases into the subsequent LLM extraction phase.

B. Phase 2: Attribute Taxonomy and Ground Truth Construction

To establish a reliable evaluation baseline for the automated extraction layer, the raw dataset (N = 250) is partitioned into a human-annotated validation subset (n = 100) and an automated inference subset (n = 150).

Prior to executing the annotation protocol, it is methodologically imperative to establish a formal taxonomy of the target attributes to be extracted from the unstructured texts. We deliberately selected four core analytical dimensions, each serving a specific structural purpose for downstream labor market analysis:

- **`Job Domain:`** Categorizes the posting into distinct technological sectors. To ensure analytical consistency, this attribute is strictly classified into seven predefined macro-domains: *Software Development, Data & AI, Infrastructure, Testing & Quality, Management & Analysis, Design & UX*, and *Others*. *Justification:* The IT market is highly heterogeneous. Aggregating data without domain stratification would introduce significant variance in downstream salary analyses; thus, this label is crucial for localized grouping and domain-specific imputation later in the pipeline.
- **`Min years of exp:`** Extracts the minimum years of work experience required for the role. *Justification:* Seniority is a primary determinant of compensation. Since raw JDs frequently contain highly ambiguous or range-based experience requirements (e.g., ”2-3 years” or ”Middle/Senior”), converting this into a single numeric value is required for statistical regression and missing value handling.
- **`Language Requirement:`** Identifies the explicit necessity of foreign language proficiency (predominantly English or Japanese). *Justification:* Given Vietnam’s prominent position in the global IT outsourcing landscape, bilingual capability acts as a major economic indicator and a notable factor associated with higher salary ranges.
- **`Skills:`** Captures the explicit technological stack requested by the employer. Instead of open-ended free-text extraction, the identified keywords are systematically mapped against a predefined, standardized taxonomy consisting of 78 core IT skills. This taxonomy was manually curated by the research team based on recurring skill mentions observed across the collected job postings. *Justification:* While this 78-skill dictionary may contain minor omissions due to the rapid evolution of niche frameworks, it successfully generalizes the general technical skill demands of the current market. Standardizing these keywords is a necessary prerequisite for constructing the skill co-occurrence matrix, enabling us to map co-occurrence patterns of technical skills across job postings.

Having defined the target taxonomy, we designed a double-blind cross-annotation protocol for the 100-sample validation subset, followed by a structured conflict resolution process. This decision was motivated by the semantic ambiguity that is common in IT job postings. For example, classifying a role as "Management & Analysis" versus "Data & AI" often depends on subtle contextual cues, and identifying implicit skill mentions requires consistent domain knowledge across annotators. Relying on a single annotator would leave individual bias and subjective interpretation unchecked, introducing noise into the reference labels and reducing the reliability of the downstream LLM evaluation.

To ensure label consistency, the two annotators completed the entire labeling process independently, without any communication or access to each other's outputs. Inter-annotator agreement for structured categorical attributes, specifically: `Job Domain`, `Min years of exp`, and `Language Requirement`, is formally quantified using Cohen's Kappa coefficient [4], defined as:

$$\kappa = \frac{p_o - p_e}{1 - p_e} \tag{1}$$

where p_o denotes the observed agreement proportion between the two annotators, and p_e denotes the expected agreement by chance, computed from the marginal label frequencies of each annotator. This formulation is preferred over simple accuracy because it corrects for the possibility that agreement may occur purely by random chance, which is a non-trivial concern in datasets with imbalanced label distributions.

For the multi-label `Skills` attribute, the extraction task is fundamentally different in nature. Each job posting may require an arbitrary number of technical skills, and the order in which they appear carries no semantic significance. Therefore, each annotator's output is treated as an unordered set of discrete technical keywords (A and B), and consistency is measured using the Jaccard similarity index [5]:

$$J(A, B) = \frac{|A \cap B|}{|A \cup B|} \tag{2}$$

This metric quantifies the proportion of skills that both annotators agree upon relative to the total number of unique skills identified by either annotator. A score of 1.0 indicates perfect agreement, while a score of 0.0 indicates no overlap whatsoever. Jaccard similarity is particularly well-suited for this task because it is invariant to set ordering and naturally penalizes both over-extraction and under-extraction of skill tokens.

The annotation process achieved an average Cohen's Kappa above 0.87 across all categorical fields and a mean Jaccard similarity exceeding 73% for tokenized skill sets, both of which indicate substantial inter-annotator agreement. For the remaining cases where discrepancies occurred—primarily driven by boundary disagreements in experience level classification or missing skill tokens—a formal consensus protocol was invoked. The two annotators jointly reviewed the original source text to resolve each conflict through discussion, producing the finalized and verified reference dataset.

C. Phase 3: Generative AI Integration and Prompt Engineering

To scale the structuring process to the remaining unannotated subset (n = 150) without incurring excessive human labor costs, we integrate a Large Language Model (LLM) as an automated information extraction engine. Specifically, we deploy the Gemini 3.1 Flash Lite model configured at Temperature = 0.0 to ensure fully deterministic outputs and strict reproducibility across all execution runs.

The extraction process is organized into two sequential stages:

1) **Calibration and Evaluation:** Before processing the full inference subset, the model is evaluated on the 100-sample Ground Truth using zero-shot prompting. Zero-shot prompting is adopted deliberately, as it assesses the model's ability to generalize from the prompt design alone, without relying on labeled examples. The model's outputs are then statistically compared against the human-verified Ground Truth to confirm extraction quality. The results indicate strong agreement: a Cohen's Kappa of 0.9371 for Job Domain, 0.8366 for Experience requirements, and 0.8402 for Language requirements. The multi-label Skill extraction achieved a Jaccard similarity of 76.35%. These results confirm that the prompt is sufficiently calibrated for large-scale inference.
2) **Inference and Merging:** Once the extraction quality is confirmed, the same prompt is applied to the remaining 150 unannotated records. The resulting outputs are standardized and seamlessly merged with the human-annotated subset to form a unified dataset of 250 records.

A key design component of this phase is the *Strict Definitions Framework* embedded within the system prompt, which is intended to prevent label inflation. While technical keywords such as *Java* or *Python* have clear and unambiguous meanings, abstract functional roles such as *Business Analysis* or *Monitoring* are often mentioned in Job Descriptions (JDs) as general responsibilities rather than explicit requirements. Without clear constraints, LLMs tend to over-extract these as core skills, artificially increasing false positives in the output.

As illustrated in Fig. 2, the prompt provides explicit definitions for each ambiguous category, limiting extraction to cases that strictly meet specific conditions. This practical approach reduces inconsistent labeling and significantly improves the overall reliability of the automated extraction layer.

D. Phase 4: Data Cleaning and Statistical Imputation

The consolidated raw dataset contains several structural attributes that require cleaning before analysis. Both numerical and categorical fields are affected by formatting inconsistencies, missing values, and outliers. To prepare the dataset for exploratory analysis, we apply a sequential cleaning and imputation pipeline.

1) Location Standardization and Experience Imputation: Free-text geographical entries are first mapped to Vietnam's 63 official administrative provinces, resolving inconsistent

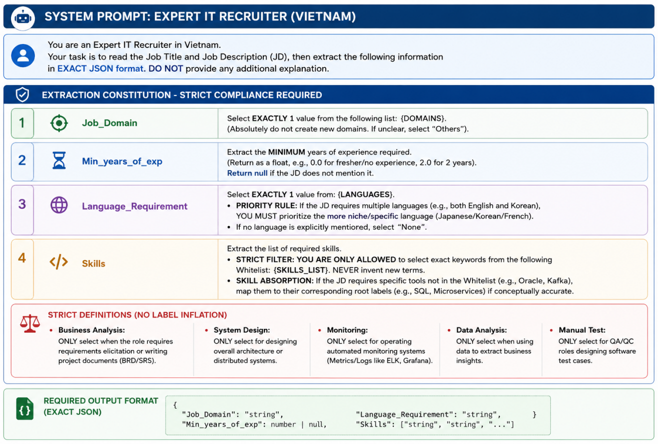


Fig. 2: Prompt Engineering architecture deployed for zero-shot information extraction using the Gemini 3.1 Flash Lite model, incorporating the extraction constitution, skill whitelist constraints, and label inflation prevention rules.

city and district naming across postings. For the experience field (`Min_years_of_exp`), a portion of records contain missing values where employers did not specify a minimum requirement. Rather than discarding these records, missing values are filled using the median experience of other postings within the same job domain. This ensures that each incomplete record receives a realistic estimate based on the standards of its specific sector.

2) Salary Parsing and Outlier Handling: Raw salary strings are parsed to extract numeric boundaries. Foreign currency values (primarily USD) are converted to VND applying a standardized market exchange rate of approximately 25,000 VND per USD, while non-numeric disclosures such as "Negotiable" or "Competitive" are treated as missing values (`NaN`). This produces two continuous variables: `Salary_Min_VND` and `Salary_Max_VND`.

Before imputing missing salary values, we first validate the integrity of available records through a two-step process. Records where the parsed minimum salary exceeds the maximum salary are identified as data entry errors and removed. Next, to reduce the influence of extreme compensation outliers on imputation, we construct separate reference pools for the minimum and maximum salary boundaries. Within each pool, records are grouped by job domain. For any domain containing at least three records, the 80th percentile ($Q_{0.80}$) is computed and used as an upper threshold. Salary values exceeding this domain-specific ceiling are excluded from the imputation reference pools.

3) Missingness Mechanism and Imputation Strategy: The parsing stage reveals a high proportion of missing values in the salary fields, which is a common characteristic of recruitment datasets in Vietnam. Simply removing incomplete records is not a viable solution, as it would reduce the sample size and introduce selection bias, given that competitive roles often withhold salary information intentionally.

We therefore model the missing data under the *Missing at Random (MAR)* assumption. MCAR is ruled out because missingness is not evenly distributed across the dataset. MNAR is also rejected because the decision to hide salary is driven by job category and seniority level, not by the salary value itself [7]. Although Multiple Imputation by Chained Equations (MICE) is a common choice for MAR data, it is not well-suited here. MICE builds iterative regression models across all features, which tends to perform poorly on small datasets ($N = 250$) with sparse categorical variables such as the 78-skill taxonomy [8].

Under the MAR assumption, the probability of a salary value being missing depends on other observed variables rather than the missing value itself. In this context, salary disclosure is a deliberate employer decision that varies by job domain and required experience level. Employers in specialized or senior roles frequently opt not to disclose compensation to preserve negotiation flexibility.

Based on this dependency structure, we implement an *Experience-Aware Nearest Neighbor* imputation strategy. For each record with a missing salary, the algorithm searches within the same job domain and applies the following tiered logic:

1) **Primary Neighborhood:** All peer records within a ± 1 year experience window are pooled. A one-year range is chosen because minor experience differences rarely shift salary brackets significantly, and pooling these records produces a more stable local sample.
2) **Fallback Neighborhood:** If no peers exist within the one-year window, the algorithm selects the record with the closest experience value as the reference point.

The missing value is then filled using the median of the identified peer group. If an entire domain has no reference data available, the overall market median is used as a final fallback.

4) Choosing the Central Tendency Metric: Mean vs. Median: A practical consideration in this imputation design is the choice between the arithmetic Mean and the Median to represent the central tendency of each peer group.

As shown in Table 2, the Mean and Median values are closely aligned across all job domains, which confirms that the 80th-percentile outlier filter applied earlier successfully normalized the salary distribution within each group.

Despite their similarity, we explicitly choose the **Median** for two reasons:

- **Small-Sample Stability:** Because peer groups are filtered by both job domain and a narrow experience window, the number of eligible records can be very small ($N \leq 3$). In such cases, the Mean is easily influenced by a single data point, whereas the Median remains stable by identifying the middle value directly.
- **Robustness to Skew:** IT salary distributions are often asymmetric even after filtering. The Median does not assume a normal distribution, making it more reliable than parametric alternatives such as MICE in this setting [8].

Job Domain	Calculated Mean (VND)	Chosen Median (VND)
Data & AI	33,411,290	**33,467,741**
Design & UX	30,683,333	**28,572,222**
Infrastructure	34,692,154	**34,093,085**
Management & Analysis	31,874,325	**31,821,052**
Software Development	36,026,303	**35,614,224**
Testing & Quality	25,111,781	**25,129,310**

TABLE 2: Comparison of Mean and Median salary values across job domains after outlier filtering.

After all missing values are filled, numerical fields are rounded to produce the final cleaned dataset, ready for downstream analysis.

IV. EXPLORATORY DATA ANALYSIS AND MARKET INSIGHTS

Following the data structuring, cleaning, and imputation pipeline, the raw job postings have been transformed into a standardized dataset of 250 records. In this section, we conduct an Exploratory Data Analysis (EDA) to identify key trends in the Vietnamese IT labor market across three dimensions: geographical distribution, compensation benchmarks, and technical skill co-occurrence patterns.

A. Job Market Landscape: Domain and Geographic Distribution

We begin by examining the distribution of job postings across domains and locations to establish a baseline understanding of the market.

Geographically, the Vietnamese IT labor market is heavily concentrated in two cities. As shown in Fig. 3a, the majority of job opportunities are located in Ho Chi Minh City and Hanoi, reflecting the clustering of technology companies, foreign investment, and skilled talent in these urban centers. Opportunities outside these two cities, including Da Nang and remote positions, represent a much smaller share, suggesting that physical presence in major tech hubs remains a common requirement for many employers in Vietnam.

From a sector perspective (Fig. 3b), *Software Development* accounts for the largest share of postings, consistent with Vietnam's established role as a regional hub for software outsourcing. At the same time, *Data & AI* and *Management & Analysis* also show notable demand, indicating a gradual shift toward data-driven and consulting-oriented roles. Foundational sectors such as *Testing & Quality* and *Infrastructure* maintain a smaller but steady presence, reflecting their supporting role in the broader software development process.

B. Salary Benchmarks: Compensation Across Domains

To understand how different IT sectors are valued in the market, we analyze the imputed `Salary_Min_VND` and `Salary_Max_VND` fields. Median salary ranges are used rather than means, as they better capture the realistic earning potential of a typical candidate while limiting the influence of extreme compensation values at the upper end of each domain.

As shown in Fig. 4, compensation varies considerably across domains. Roles in *Data & AI*, *Management & Analysis*, and *Software Development* offer the highest salary ranges, with maximum salaries reaching 45 million VND per month. This reflects the strong demand and relatively limited supply of professionals in these areas. *Infrastructure* also displays a notably wide salary band, driven by the specialized nature of cloud and systems engineering work.

In contrast, *Design & UX* and *Testing & Quality* show more modest compensation. While both domains play important roles in the software development process, their skill sets are more standardized across the market, resulting in narrower salary ranges. The localized imputation methodology applied in Phase 4 ensures that these figures reflect the realistic compensation levels of each domain without cross-sector distortion.

C. Technological Skill Demand: Top Competencies by Domain

Analyzing skill requirements at the domain level provides useful guidance for both job seekers and educational institutions. By examining the tokenized `Skills` attribute, we identified the top 5 most frequently requested skills within each job domain, expressed as a percentage of postings within that domain, as shown in Fig. 5.

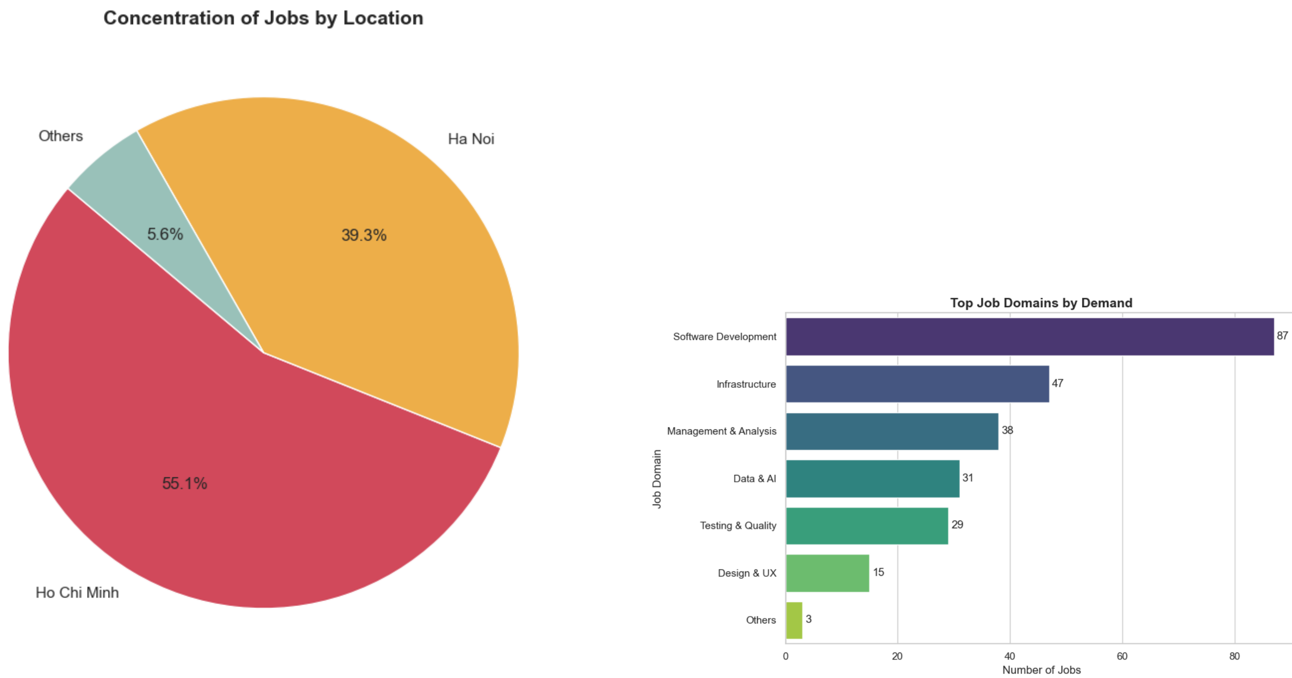
In the *Data & AI* domain, **Python** (67.7%) and **SQL** (54.8%) are the two most requested skills by a clear margin, followed by **Data Engineering** (45.2%), **API** (41.9%), and **Data Warehousing** (41.9%). This distribution reflects the end-to-end nature of data roles, where professionals are expected to handle everything from raw data processing to model deployment and database querying.

The *Software Development* domain is led by **API** (56.3%), **CI/CD** (43.7%), and **Git** (40.2%), alongside **AWS** (35.6%) and **Java** (34.5%). The prominence of DevOps-adjacent skills such as CI/CD and Git alongside core programming languages suggests that modern software engineering roles in Vietnam increasingly expect candidates to be familiar with deployment and version control practices, not just application development.

In *Infrastructure*, the top skills are **Security** (76.1%), **Monitoring** (71.7%), and **Networking** (60.9%), with **Linux** (50.0%) and **AWS** (47.8%) also appearing frequently. This reflects the operational focus of infrastructure roles, where system reliability, network management, and cloud platforms are the primary concerns.

For *Management Analysis*, **Business Analysis** (64.9%) and **Scrum / Agile** (51.4%) dominate, indicating that these roles prioritize project coordination and requirements gathering over technical implementation. **API**, **Data Analysis**, and **SQL** each appear in 24.3% of postings, suggesting that a basic level of technical literacy is still expected.

The *Design UX* domain is unsurprisingly led by **Design**, appearing in all 100% of postings, followed by **Data Analysis** and **Frontend Basics** (40.0% each). The presence of Data



(a) Job postings by location

(b) Job postings by domain

Fig. 3: Distribution of IT job postings by administrative location (a) and technological domain (b) in Vietnam.

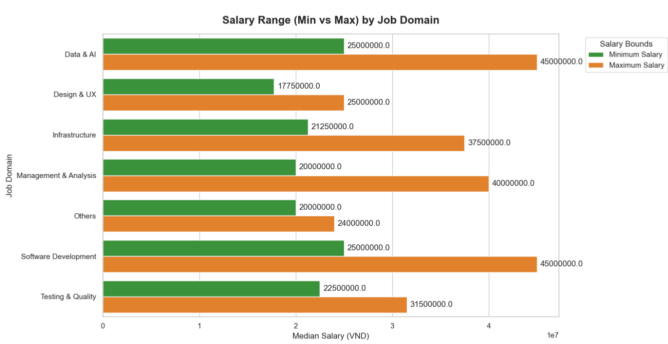


Fig. 4: Median minimum and maximum salary boundaries (VND) across IT job domains.

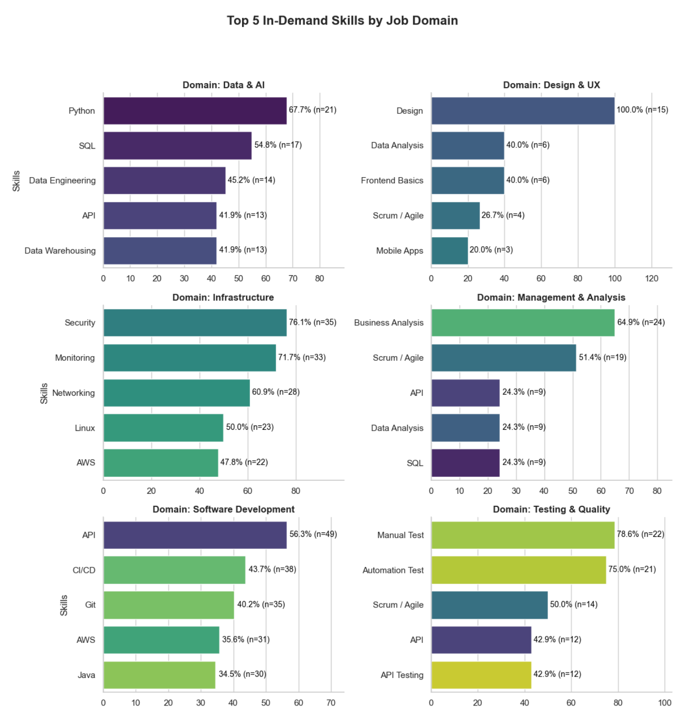


Fig. 5: Top 5 most frequently requested skills across six IT job domains.

Analysis here indicates that UX roles in Vietnam are increasingly expected to support design decisions with user data.

Finally, *Testing Quality* roles are split between **Manual Test** (78.6%) and **Automation Test** (75.0%), showing that both testing approaches remain equally relevant in the current market. **Scrum / Agile** (50.0%), **API** (42.9%), and **API Testing** (42.9%) round out the top five, reflecting the integration of QA within agile development workflows.

D. The Economic Value of Foreign Languages

Given Vietnam's position as a regional IT outsourcing hub, we examine whether foreign language requirements are associated with higher compensation. The dataset is divided into two groups: roles that explicitly require a foreign language and roles that do not.

As shown in Fig. 6, the two groups share the same median salary of **30 million VND per month**. However, roles requiring a foreign language display a noticeably wider salary distribution, with a higher upper quartile and more frequent high-salary outliers reaching up to 100 million VND. This suggests that while language requirements do not consistently raise baseline compensation, they are associated with higher earning potential at the senior or client-facing end of the market. In the context of Vietnam's IT outsourcing industry, professionals with strong foreign language skills are often placed in international-facing roles, which tend to offer more variable and performance-linked compensation structures.

It is worth noting, however, that the overlap between the two distributions is substantial. Language proficiency alone is unlikely to be the primary driver of salary differences; domain expertise and seniority level remain the dominant factors.

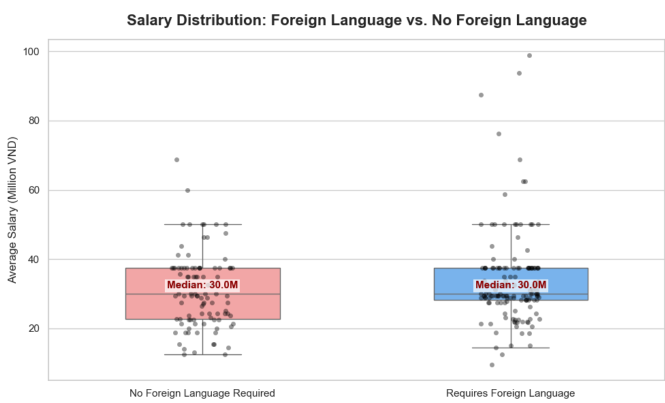


Fig. 6: Salary distribution comparison between roles with and without foreign language requirements.

E. Skill Co-occurrence Patterns

In practice, employers rarely look for a single isolated skill. Instead, job postings typically require a combination of tools and technologies that work together within a specific technical stack. To map these relationships, we constructed a skill co-occurrence matrix for the top 20 most frequently appearing skills, counting how often each pair of skills appears together within the same job posting.

As shown in Fig. 7, several strong pairings stand out. **CI/CD and AWS** show the highest co-occurrence count (44), followed closely by **CI/CD and Docker** (41) and **CI/CD and API** (41). This cluster indicates that DevOps-related skills are tightly bundled in the market, where proficiency in one tool is almost always expected alongside the others.

AWS and Azure also co-occur frequently (36), suggesting that multi-cloud familiarity is increasingly valued. Meanwhile, **Microservices and API** (35) and **Security and Monitoring** (30) represent two other well-defined skill bundles, reflecting the architectural patterns commonly found in enterprise software and infrastructure roles respectively.

In contrast, **Business Analysis** shows consistently low co-occurrence with most technical skills, which is expected given its non-technical nature. **SQL** maintains moderate co-occurrence across a wide range of skills including Python, Java, and API, reinforcing its role as a foundational cross-domain skill that appears regardless of the primary technology stack.

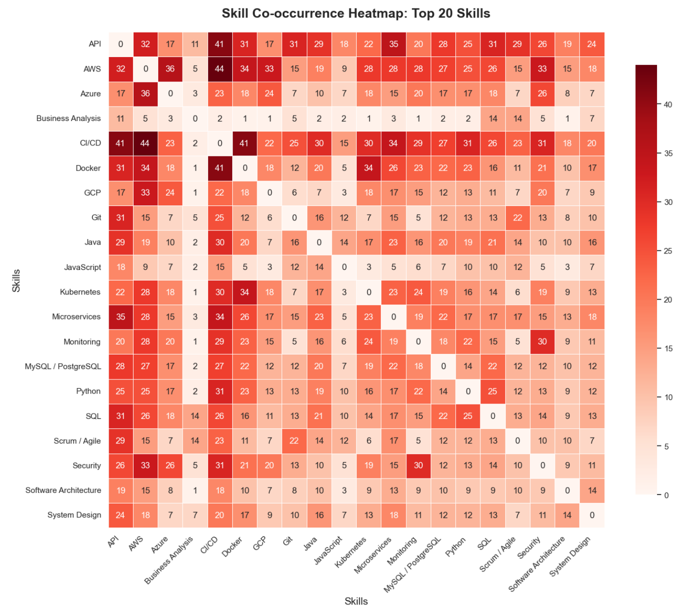


Fig. 7: Co-occurrence frequency matrix of the top 20 technical skills across all job postings.

V. CONCLUSION

In this study, we developed an end-to-end pipeline to extract, clean, and analyze unstructured IT job postings from the Vietnamese labor market. By deploying the Gemini 3.1 Flash Lite model in fully deterministic mode and designing a structured prompt constitution, we achieved consistent and reliable information extraction across all four target attributes. The pipeline was validated against a human-annotated Ground Truth, yielding a Cohen's Kappa above 0.83 for categorical fields and a Jaccard similarity above 76% for skill extraction, suggesting that the automated extraction layer achieves a level of consistency approaching human annotation on the validation subset.

To address the high proportion of missing salary data, we modeled the missingness under the Missing at Random (MAR) assumption and implemented an Experience-Aware Nearest Neighbor imputation strategy. By grouping peer records within the same job domain and a narrow experience window, the algorithm produces salary estimates that reflect realistic market conditions without relying on global statistics or complex parametric models.

The subsequent exploratory analysis revealed several structural patterns in the Vietnamese IT labor market. Job opportunities remain heavily concentrated in Ho Chi Minh City and Hanoi. Compensation varies considerably across domains, with Data & AI, Software Development, and Management &

Analysis offering the widest salary ranges. Foreign language requirements are associated with greater salary variability at the upper end of the market, though the median compensation remains similar between the two groups. Skill co-occurrence analysis further reveals that DevOps-related tools such as CI/CD, Docker, and AWS form a tightly coupled skill bundle, while SQL maintains broad relevance across multiple domains.

Overall, this work delivers a structured, analysis-ready dataset and a documented preprocessing methodology that can serve as a reproducible foundation for future labor market research in the Vietnamese IT sector.

VI. Limitations

While this study presents a structured and reproducible pipeline for IT recruitment data analysis, several limitations should be acknowledged.

Dataset Scale. The final dataset consists of 250 job postings, which is sufficient for exploratory analysis but may not fully represent the breadth of the Vietnamese IT labor market. Findings related to salary benchmarks and skill demand should be interpreted as indicative trends rather than definitive market statistics.

Salary Data Availability. Approximately 75% of collected job postings did not disclose explicit salary figures, using non-numeric responses such as "Negotiable" or "Competitive" instead. Although the Experience-Aware Nearest Neighbor imputation strategy produces statistically reasonable estimates, the high proportion of imputed values means that salary-related findings carry inherent uncertainty. All salary benchmarks reported in this study should be understood as estimated distributions rather than ground-truth market figures.

Skill Taxonomy Coverage. The 78-skill whitelist was manually curated based on recurring mentions observed in the collected postings. While this approach ensures standardization, it may fail to capture emerging or niche technologies that were underrepresented in the sample. As the IT landscape evolves rapidly, the taxonomy may require periodic updates to remain current.

Temporal Scope. The dataset reflects a static snapshot of the Vietnamese IT job market at the time of collection. Labor market conditions, salary levels, and skill demands change over time, and the findings may not generalize to different time periods without re-collection and re-processing.

Platform Coverage. Data was collected exclusively from two platforms, ITviec and TopDev, which primarily serve the formal IT sector. Job postings from smaller platforms, social media, or informal recruitment channels are not represented, which may introduce a degree of selection bias toward larger companies and more structured roles.

LLM Extraction Constraints. Although the Gemini 3.1 Flash Lite model achieved strong agreement metrics against the human Ground Truth, zero-shot extraction is not infallible. Edge cases involving highly ambiguous job descriptions or domain-specific terminology may still result in minor labeling errors that are difficult to detect without full manual verification of all 150 inferred records.

Appendix A
Datasheets for Datasets

This appendix provides a standardized Datasheet for the constructed IT Recruitment dataset, following the framework proposed by Gebru et al. [9], to ensure transparency, accountability, and reproducibility.

A. Motivation

For what purpose was the dataset created? The dataset was created to support research on the Vietnamese IT labor market, addressing the limited availability of structured data for studying technical skill demands and salary distributions in this domain.

Who created the dataset? The dataset was constructed by a two-member student research group as part of an academic course project, using a semi-automated pipeline combining human annotation and LLM-assisted extraction.

B. Composition

What do the instances that comprise the dataset represent? Each instance represents a single IT job posting collected from the Vietnamese job market.

How many instances are there? The dataset consists of 250 records.

Is the dataset a sample from a larger set? Yes. The 250 records represent a non-random sample of IT job postings available on ITviec and TopDev during the collection period. The sample is not guaranteed to be statistically representative of the full Vietnamese IT labor market.

What data does each instance consist of? Each record contains structured attributes including Job Title, Job Domain, Company, Standardized Location, Minimum Years of Experience, Language Requirements, Technical Skills, and Monthly Salary Boundaries (Min/Max in VND).

Is any information missing from individual instances? Yes. Approximately 75% of records did not disclose explicit salary figures. These missing values were handled through statistical imputation as described in the preprocessing pipeline.

C. Collection Process

How was the data associated with each instance acquired? Data was collected semi-manually via direct browser interaction on ITviec and TopDev, capturing rendered page content to preserve the original plain-text structure of each posting.

What mechanisms were used to collect the data? No automated scrapers were used. Data was captured directly from the browser-rendered output to avoid anti-bot restrictions and preserve formatting that would otherwise be lost during HTML parsing.

Over what timeframe was the data collected? Data collection was conducted in May 2026.

Who was involved in the data collection process? Both members of the student research group participated equally in data collection as part of their coursework. No external compensation was involved.

Does the dataset strictly preserve the original state? Yes. The initial collection adhered to the *Immutable Raw Principle*, preserving all original formatting, line breaks, and characters prior to preprocessing.

D. Preprocessing, Cleaning, and Labeling

Was there any preprocessing or data cleaning? Yes. Three main cleaning steps were applied: free-text locations were mapped to Vietnam's 63 standard administrative provinces; raw salary strings were parsed into structured numeric boundaries; and missing values for salary and experience were handled via a Groupby Median imputation strategy under the Missing at Random (MAR) assumption.
Was the raw data saved in addition to the processed data? Yes. The original uncleaned dataset was preserved separately in accordance with the Immutable Raw Principle, and is retained alongside the final processed version.
How was the data labeled? A 100-sample subset was manually labeled through independent cross-annotation by both team members, followed by a structured conflict resolution process. The remaining 150 records were automatically labeled via zero-shot prompting using Gemini 3.1 Flash Lite at Temperature = 0.0.

E. Uses

Has the dataset been used for any tasks already? Yes. The dataset was used to conduct an Exploratory Data Analysis (EDA), examining salary distributions by domain, skill demand by sector, and skill co-occurrence patterns across job postings.
What tasks could the dataset be used for in the future? The dataset could support predictive salary modeling, skill gap analysis, automated career guidance systems, and longitudinal labor market tracking.
Are there tasks for which the dataset should not be used? The dataset should not be used to make consequential decisions about individual job applicants or employers, as it represents a small, non-random sample with a high proportion of imputed salary values.

F. Distribution and Maintenance

Will the dataset be distributed to third parties? Yes. The final processed dataset, the Ground Truth subset, and the full preprocessing codebase (Jupyter Notebooks) are made available to support reproducibility and future research.
How will the dataset be distributed? The dataset is distributed in CSV format alongside the preprocessing notebooks via the project repository.
Will the dataset be distributed under a license? The dataset is intended for academic and non-commercial research use only. Users are requested to cite this paper if the dataset is used in subsequent work.
Who will be supporting/hosting/maintaining the dataset? The authors of this paper will maintain the dataset repository and respond to questions regarding its use and structure.

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